

How to build self-explaining fuzzy systems: From interpretability to explainability

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Abstract—Fuzzy systems are known to provide not only accurate but also interpretable predictions. However, their explainability may be undermined if non-semantically grounded linguistic terms are used. Additional non-trivial challenges would arise if a prediction were to be explained counterfactually, i.e., in terms of hypothetical, non-predicted outputs. In this paper, we explore how both factual and counterfactual automated explanations can justify the output of fuzzy rule-based classifiers, and thus contribute to making them more trustworthy. Moreover, we demonstrate how end user preferences can be handled by customizing automated explanations, making them interactive, personalized, and therefore, human-centric. The full immersive article at IEEE Xplore provides detailed interactive examples for better understanding.

I. INTRODUCTION

THE rapid ongoing evolution of artificial intelligence (AI) demands explainability at the level of data, algorithms, models, systems, and predictions, as a prerequisite for Explainable and Trustworthy AI [1], [2]. Whereas black-box classifiers can generate accurate but poorly explainable predictions, a fuzzy rule-based classification system (FRBCS) can produce not only accurate but also human-friendly and interpretable predictions [3], [4]. On one hand, a prediction of an FRBCS can be explained factually (e.g., answering the why-question “Why is the test instance predicted to be of class P ?”) by listing the feature-value pairs that appear in the antecedent of the rule that has the highest activation degree. On the other hand, the given prediction can be explained counterfactually (e.g., answering the why-not-question “Why is the test instance not predicted to be of class Q ?”). Such so-called counterfactual explanations presuppose minimal changes in the feature values of the test instance for the prediction to change, making it maximally relevant to the test instance under consideration [5].

One of the major limitations of static responses to why-not-questions is a lack of interaction with the explainee. Instead, it is desired that the end user can customize the given explanation to further increase its utility. To do so, we devised a method for generating explanations, which incorporates user preferences to produce tailored explanations, allowing for acceptable ranges within the feature space where feasible counterfactuals can be located. As the method generates single and valid counterfactuals that are minimally different from the test instance, it allows the end user to draw difference between

linguistic explanations for fuzzy classifiers and those for other supervised machine learning-based classifiers.

In this work, a use case on leaf classification is considered. Imagine one has a leaf and intends to use a fuzzy logic-based classifier to categorize it based on its geometric properties (e.g., Area and Perimeter). The original FLAVIA dataset [6] contains a collection of nearly 1800 images of leaves of 33 different classes found in the region of Yangtze Delta, China. For illustrative purposes, a subset of only 310 leaves is considered. The leaves are taken from the original dataset and grouped into five classes. Thus, the leaf under consideration is said to belong to one of the following classes in accordance with their scientific names: *Aesculus chinensis*, *Berberis anhwienensis*, *Cercis chinensis*, *Phoebe zhennan*, or *Lagerstroemia indica*.

Even if the classifier were interpretable, without the assistance of a botanist, it may be difficult to judge whether the classifier categorized the leaves properly. It is therefore necessary to enhance the classifier with an explanation generation module so that it can automatically explain its own prediction and go a step ahead of interpretability (i.e., the user executes the interpretation) to explainability (i.e., the classifier processes the interpretation and provides the end user with a tailored meaningful explanation).

In the rest of this manuscript and the related immersive article, the readers will be presented with a practical, step-by-step guide on how to (1) build interpretable FRBCSs, (2) add a linguistic layer on top of approximate fuzzy classifiers (i.e., those with local semantics), (3) generate factual and counterfactual explanations for the resulting systems, and (4) yield user-preferred counterfactuals.

II. EXPLAINABLE FUZZY SYSTEMS

An explanation is a presentation of the reasoning, functioning and/or behavior of an explicandum (e.g., an intelligent system) in human-understandable terms, to be conveyed from the explainer to the explainee. Imposing structural and semantic constraints is required to ensure, by design, interpretability of FRBCSs, and this is a prerequisite for endowing fuzzy classifiers with explainability and trustworthiness [4]. Given an FRBCS, understanding the following concepts is essential to generate explanations for the fuzzy classifier’s predictions:

- *Linguistic Variables*. The user must define the meaning of linguistic terms to be used in the explanations. The use of strong fuzzy partitions is mandatory for the sake of interpretability.

- *Linguistic approximation.* In case of dealing with approximate fuzzy rules which lack linguistic interpretability (e.g., “IF Perimeter=MF0 AND Area=MF1 THEN Leaf IS Chinese horse chestnut”), they should be mapped to expert-annotated terms (e.g., MF0=Small for Perimeter and MF1=Very small for Area).
- *Surface realization.* The most relevant piece of information is verbalized using a template-based natural language generation pipeline [7]. Given a test instance, the rule above can be transformed into an explanation in natural language as follows: “The leaf is Chinese horse chestnut because Perimeter is small and Area is very small.”

Being aware of the concepts presented above, it is then possible to apply them to generate factual and counterfactual explanations as follows [8]:

- *Factual explanation generation.* The IF-THEN rule with the highest activation degree is linguistically approximated (if necessary) and transformed into a well-formed sentence in natural language.
- *Counterfactual explanation generation.* All rules leading to the given counterfactual class are vectorized in terms of all possible feature-value pairs. Similarly, the test instance vector is composed of membership function values for all the linguistic terms for all the features. The relevance of each counterfactual rule is estimated by calculating the Euclidean distance between the test instance vector and all the counterfactual rule vectors. Then, the rule vector with the minimal distance from the test instance is subsequently transformed into a sentence in natural language.

Alternatively, counterfactuals can also point to individual data points that are minimally different from the test instance under consideration. The user preference-based methodology to generate counterfactual explanations is as follows [9]:

- *User preferences and nearest neighborhood for counterfactual explanation generation.* The first step involves asking the user to furnish a list of editable features and their respective boundary ranges in the input sample (feature space). Then, this input is utilized to demarcate a neighborhood space for exploring counterfactuals. Then, perturbations in the actual input adhering to user preferences are performed to generate candidate counterfactuals as a synthetic dataset. Heuristics set rules for initial perturbation of features based on strong positive correlations, followed by an exhaustive search to identify candidate counterfactuals. For example, to find the counterfactual outcome for the given test instance, the user is supposed to specify affordable ranges for Area and Perimeter. The subspace adhering to the affordable ranges is exploited to find the candidate counterfactuals that guarantee the desired outcome. To calculate and reduce the distance between the test instance and the candidate counterfactuals, the method pioneered by Wachter *et al* [10] is applied. Our approach iteratively seeks to identify better counterfactuals adhering to the user affordable ranges.

III. CONCLUSION

Fuzzy sets and systems are well known because of their ability to deal with approximate reasoning, mainly in control applications. It is timely to explore their ability to deal with linguistic variables and rules in the context of Explainable and Trustworthy AI. Even though great effort in the research community has been devoted to establishing the basis for ensuring interpretability of fuzzy systems, more work is still required for ensuring explainability. In this work, we have sketched how to make fuzzy classifiers not only interpretable by careful design but also explainable. Explainable Fuzzy Systems can provide users with both factual and counterfactual explanations in natural language. More details and examples can be found in the online full paper at IEEE Xplore.

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